LESSON PLAN

EWIS (21)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Purpose and technical details of VOR/ILS/ MKR system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Purpose and technical details of VOR/ILS/ MKR system
5. Specific Objectives	21.1.1.1 Trainee will be able to understand the basic theory of communication.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of VOR/ILS/MKR schematic diagrams from IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of ground-based navigation aids?
(b) Name three types of radio navigation systems used in aviation.
(c) Define the term 'VOR' in the context of navigation.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic theory of radio communication • VOR system purpose and technical details • ILS system purpose and technical details • MKR system purpose and technical details	L & Ds	PPT, IB	15 min
(b)	• Operational principles of VOR signals • ILS localizer and glide slope functions • Marker beacon signal types and indications	L & Ds	PPT, PDF	20 min
(c)	• Comparison of VOR, ILS, and MKR accuracy • Integration of systems in flight navigation • Troubleshooting common navigation errors	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the VOR system determines aircraft position relative to the station.
(b) Describe the role of the glide slope in the ILS system during approach.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of VOR and ILS will be applied in the upcoming lesson on sonar systems (LP 05) regarding underwater navigation analogies.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to sonar system. Purpose and technical details of active and passive sonar system.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min